Deploy a Web Application to Kubernetes using minikube

Step 1 - Initialize minikube

- Run *minikube start* and *minikube status* to start and check the stratus of minikube

```
┌(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└(09:46:25)──> minikube start                                                ──(Fri,Dec12)─┘
😄  minikube v1.37.0 on Debian bookworm/sid
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🏃  Updating the running docker "minikube" container ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
┌(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└(09:47:25)──> minikube status                                               ──(Fri,Dec12)─┘
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 2 - Write the deployment YAML file

```
┌(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└(09:43:07)──> cat deployment.yaml                                           ──(Fri,Dec12)─┘
apiVersion: apps/v1
kind:Deployment
metadata:
  name:nginx-deployment
spec:
  selector:
    matchLabels:
      app:nginx
  template:
    metadata:
      labels:
        app:nginx
    specs:
      container:
        - name:nginx
          image:nginx:latest
          ports:
            -containerPort:80
```

Step 3 - Apply deployment file and verify

```
┌─(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(09:45:29)──> kubectl apply -f deployment.yaml                             ──(Fri,Dec12)─┘
deployment.apps/nginx-deployment created
┌─(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(09:45:31)──> kubectl get deployments                                      ──(Fri,Dec12)─┘
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   1/1     1            1           11s
```

Step 4 - Expose a port to access the nginx deployment and verify

```
┌─(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(09:46:04)──> kubectl expose deployment nginx-deployment --type=NodePort --port=80 ──(Fri,Dec12)─┘
service/nginx-deployment exposed
┌─(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(09:46:20)──> kubectl get svc                                              ──(Fri,Dec12)─┘
NAME               TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes         ClusterIP   10.96.0.1      <none>        443/TCP        11h
nginx-deployment   NodePort    10.107.6.13    <none>        80:31968/TCP   6s
```

Step 5 - Run the service using minikube

```
┌─(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(09:47:33)──> minikube service nginx-deployment                            ──(Fri,Dec12)─┘
|-----------|------------------|-------------|---------------------------|
| NAMESPACE |       NAME       | TARGET PORT |            URL            |
|-----------|------------------|-------------|---------------------------|
| default   | nginx-deployment | 80          | http://192.168.49.2:31968 |
|-----------|------------------|-------------|---------------------------|
🎉  Opening service default/nginx-deployment in default browser...
┌─(~/Desktop/devops)──────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(09:47:46)──> Opening in existing browser session.                         ──(Fri,Dec12)─┘
```

Output -